Convex Optimization

① Hessian Matrix

It consists of all second order partial derivatives of a function.

For two variables:

$$z = f(x,y)$$

$$H_1 = \begin{array}{c c} & \begin{array}{cc} f_x & f_y \end{array} \\ \begin{array}{c} f_x \\ f_y \end{array} & \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \end{array} \quad \because f_{xy} = f_{yx}$$

For three variables

$$w = f(x,y,z)$$

$$H_2 = \begin{array}{c c} & \begin{array}{ccc} f_x & f_y & f_z \end{array} \\ \begin{array}{c} f_x \\ f_y \\ f_z \end{array} & \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} \end{array} \quad \because \begin{array}{l} f_{xy} = f_{yx} \\ f_{xz} = f_{zx} \\ f_{zy} = f_{yz} \end{array}$$

② Convex Function:

If Hessian matrix is positive definite then function is convex.

Remarks: If function is convex then critical point is global/absolute minima.

Positive Definite

For two variables: $z = f(x,y)$

$H_1 = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$, If $D_1 = f_{xx} > 0$, $D_2 = \text{determinant}(H_1) = |H_1| > 0$ $\Rightarrow H_1$ is positive definite $\Rightarrow$ function is convex.

For three variables: $w = f(x,y,z)$

$H_2 = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix}$, If $D_1 = f_{xx} > 0$, $D_2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0$, $D_3 = \text{determinant}(H_2) = |H_2| > 0 \Rightarrow H_2$ is positive definite. $\Rightarrow$ function is convex.

Concave Function:

If Hessian matrix is negative definite then function is concave.

Remarks: If function is concave then critical point is global/absolute maximum point.

Negative Definite

For two variables : $z = f(x,y)$

$$H_1 = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}, \text{ If } D_1 = f_{xx} < 0, \; D_2 = |H_1| > 0$$

$\Rightarrow$ H_1 is negative definite

$\Rightarrow$ function is concave.

For three variables : $w = f(x,y,z)$

$$H_2 = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix}, \text{ If } D_1 = f_{xx} < 0, \; D_2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0$$

and $D_3 = |H_2| < 0$

$\Rightarrow$ H_2 is negative definite.

$\Rightarrow$ function is concave.